

Project Pitch: Indoor Cleaning Robot (Voom-Bot)

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Target Specialisation: Option B - Autonomous Robotics (Webots)

Hardware Platform: Pioneer 3-DX

Executive Summary

Voom-Bot is an autonomous mobile robot designed to navigate indoor environments such as classrooms, libraries and living spaces while performing routine cleaning. The project focuses on perception driven navigation, where the robot explores an unfamiliar environment, building spatial awareness through grid based mapping, avoiding obstacles and performing cleaning tasks safely.

The robot operates fully autonomously using a combination of sensor inputs and structured decision making to ensure both efficiency and safety for users and surrounding objects.

System Architecture (Sense–Think–Act)

The Pioneer 3-DX model in Webots provides a stable and realistic platform for indoor navigation and control.

- **Sensing (Inputs):**
 1. **Distance Sensors (Sonar/Proximity):** Multiple sensors provide 360° proximity data, enabling reliable obstacle detection and collision avoidance.
 2. **Camera:** Used to detect basic visual cues such as contrast or obstacles to guide movement.
 3. **Microphones:** The omnidirectional microphone detects simple audio triggers (e.g., a clap) to start or pause the cleaning process.
- **Processing (Thinking):**
 1. **Finite State Machine (FSM - Core Logic):** This is the brain that controls different behaviours, allowing clear, structured design and analysis. It makes complex decisions to prevent collisions and prioritise safety.
 2. **Perception Module:** The Perception Module uses reactive, event-driven logic where the robot continuously monitors the environment and triggers a predefined behavioural response whenever a specific condition is detected, such as an obstacle, dead end, completed task, or human presence.
 3. **Multi-Condition Decision Logic:** The system will use multi-condition decision logic by combining sensor inputs (distance, vision, audio) with internal states such as battery level to better determine behaviour transitions. Safety conditions override all other behaviours.
- **Actuation (Outputs):**
 1. **LED Indicators:** Provide visual feedback for robot states (cleaning, paused, error).
 2. **Drive Motors:** Use wheel rotations to allow movement, which allows the robot's navigation.
 3. **Speaker:** To notify users that the robot is incapable of operating (Unsafe or Out of power).
 4. **Vacuum Attachment:** Simulates the cleaning function while the robot is active.

Behavioural Logic (FSM)

State	Trigger Condition	Action / Output
IDLE / PAUSED	Startup, human interaction, fault detected	Stops movement, LEDs off
CLEANING	Clear path, sufficient battery, and activation command	Moves forward while vacuuming
OBSTACLE AVOIDANCE	Objects detected within threshold distance	Rotates and redirects to avoid collision
DEAD END RECOVERY	No valid path OR Sensors blocked	Reverses, rotates, and finds new path
CHARGING / RETURN TO DOCK	Low Battery OR Cleaning complete	Plays sound and navigates back to the charging dock.

Specialisation: Autonomous Robotics

To meet the **Option B** requirements, Voom-Bot implements:

- **Obstacle Avoidance:** It uses Distance Sensors with set distance thresholds to avoid collisions with objects in its proximity.
- **Perception-Driven Decision:** It analyses data from the Distance Sensors, microphone, and human input to plan an efficient cleaning path.
- **Navigation:** Exploration based navigation using grid mapping to track visited vs unvisited areas and improve coverage efficiency.

Safety & Robustness

- **Human/Pet Detection:** Detects nearby movement or objects using sensors and pauses operation to prevent collision.
- **Low Battery Protection:** Triggers return to dock when battery falls below a threshold.
- **Cliff/Drop Detection:** Distance sensors and the camera detect ledges, like stairs, and stops the robot from falling.
- **Stuck Detection:** If the wheels spin without motion, the robot is set to recovery to prevent motor damage.

Advanced Component (10%)

Autonomous Navigation (A Grid-Based Mapping): Voom-Bot implements a basic grid-based mapping system that tracks visited vs unvisited areas in the environment to prevent repeated coverage of the same space and improve overall cleaning efficiency.